

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

Annexure A

Code of Conduct:

BHUVANESHWARAN J(192511402)

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

BHUVANESHWARAN J

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A manufacturing plant has collected sensor records (vibration level, temperature, operating hours, past maintenance count, load percentage) to build a predictive model for identifying machines likely to fail. The plant also wants to train an AI agent to recommend maintenance actions (Inspect / Lubricate / Replace Part / Continue Operation) based on a machine's condition over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this manufacturing dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (far fewer failure events than normal operation records) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a predictive-maintenance use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Equipment Failure Prediction

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed failures) and suggest how to reduce them - justify from a safety perspective.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment on the factory floor.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naive Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of failure from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Action Recommendation

- (a) Model the maintenance recommendation as an MDP. Define: States (machine health stages), Actions (Inspect / Lubricate / Replace Part / Continue Operation), Reward function (uptime improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 machine episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for maintenance decisions (worker safety, cost trade-offs, explainability).

Scenario: Predictive maintenance for a manufacturing plant. The source assessment asks for inductive learning, a Decision Tree, statistical learning, and Q-Learning for maintenance recommendations. The supplied assessment does not provide an actual sensor dataset or measured model results; therefore, the numerical examples below are clearly stated illustrative values suitable for demonstrating the required methods.

Task 1: Data Preparation & Inductive Learning

1(a) Inductive learning approach, target variable and features

Inductive learning learns a general rule from previously observed machine records and uses that rule to predict the outcome for a new machine condition.

Target variable: Machine failure (1 = failure likely/occurred, 0 = normal operation).

Feature	Why it is important
Vibration level	Abnormally high vibration can indicate bearing wear, imbalance, looseness or mechanical damage.
Temperature	Increasing temperature may indicate overheating, friction, lubrication problems or component degradation.
Operating hours	Higher operating hours generally increase wear and the probability of maintenance or failure.
Past maintenance count	Provides information about the machine's maintenance history and recurring problems.
Load percentage	High or changing load can increase stress and accelerate component deterioration.

1(b) Handling class imbalance

Failure events are much fewer than normal-operation records, so a model may otherwise become biased toward the majority class. **SMOTE** is a suitable primary strategy because it creates synthetic minority-class examples instead of simply duplicating existing failures. SMOTE must be applied only to the training portion after the train/test split to avoid data leakage. Class weights can also be used as an alternative.

1(c) 70:30 split and preprocessing

Use **70% training data and 30% testing data**. The larger training portion gives the model enough historical machine examples, while the 30% test set provides a meaningful independent estimate. A stratified split should preserve the failure/normal proportion.

Preprocessing: inspect and impute missing values; standardise numerical features when using Logistic Regression; encode categorical variables if present; apply SMOTE only to training data; and use the same fitted preprocessing pipeline for the test set.

Task 2: Decision Tree for Equipment Failure Prediction

2(a) Tree construction and root-node selection

A Decision Tree recursively divides the data into groups that are increasingly pure with respect to failure/non-failure. The root is selected using the feature that gives the best impurity reduction, measured by Information Gain or Gini reduction.

Illustrative root selection: Assume Vibration = 0.24, Temperature = 0.18, Load = 0.12, Operating Hours = 0.10 and Maintenance Count = 0.07. Vibration therefore becomes the root node.

Level	Illustrative decision
1 (Root)	Vibration $\leq$ threshold $\rightarrow$ lower-risk branch; Vibration $>$ threshold $\rightarrow$ higher-risk branch
2	Temperature $\leq$ threshold / $>$ threshold
3	Operating Hours or Load $\leq$ threshold / $>$ threshold

Conceptual tree: Vibration $\rightarrow$ Low $\rightarrow$ Temperature $\rightarrow$ Operating Hours $\rightarrow$ Normal/Warning; High $\rightarrow$ Temperature $\rightarrow$ Load/Operating Hours $\rightarrow$ Warning/Failure.

2(b) Model evaluation and False Negatives

The important metrics are Accuracy, Precision, Recall and F1-score. For predictive maintenance, **Recall for the failure class is especially important** because a False Negative means a machine likely to fail is incorrectly predicted as normal.

Metric	Illustrative test result	Meaning
Accuracy	91%	Overall proportion of correct predictions.
Precision	84%	Among predicted failures, proportion that are actual failures.
Recall	89%	Among actual failures, proportion correctly detected.
F1-score	86%	Balance between precision and recall.

To reduce False Negatives, increase the weight/cost of failure errors, tune the classification threshold toward higher recall, use class weights or SMOTE on training data, and tune the tree using recall or F1 rather than accuracy alone. An undetected failure can cause equipment damage, production downtime and worker-safety risks.

2(c) Post-pruning and deployment choice

Cost-complexity post-pruning removes branches that add little predictive value. It reduces overfitting and generally makes the tree simpler and easier to explain.

Model	Illustrative Accuracy	Observation
Pre-pruned tree	90%	Simple and controlled complexity, but may stop useful splits too early.
Post-pruned tree	92%	Better generalisation in this illustrative comparison and fewer unnecessary

For factory-floor deployment, the **post-pruned model** would be preferred if validation confirms that it maintains high failure recall. It is simpler, less prone to overfitting, easier for engineers to inspect, and more suitable for explainable maintenance decisions.

Task 3: Statistical Learning for Risk Stratification

3(a) Logistic Regression and comparison with the Decision Tree

Logistic Regression models the probability of failure using the input features. It outputs a probability between 0 and 1, which can be converted into a class using a chosen threshold.

Model	Illustrative Accuracy	Illustrative AUC-ROC
Decision Tree (post-pruned)	92%	0.93
Logistic Regression	90%	0.95

These numbers are illustrative because the supplied assessment contains no dataset. A statistical model is preferable when a smooth probability estimate, stable coefficients, and a relatively interpretable relationship are desired. A Decision Tree is preferable when nonlinear thresholds and interactions are important and rule-like explanations are valuable.

3(b) Feature importance analysis

Rank	Decision Tree	Logistic Regression (absolute coefficient)
1	Vibration level	Vibration level
2	Temperature	Temperature
3	Load percentage	Operating hours

The two models agree strongly on vibration and temperature in this illustrative analysis. They differ on the third predictor because Decision Trees rank features by impurity reduction, whereas Logistic Regression ranks them through fitted coefficient magnitude after scaling.

Task 4: Reinforcement Learning for Maintenance Action Recommendation

4(a) MDP formulation

Component	Definition
States	S0 = Healthy, S1 = Warning, S2 = High Risk, S3 = Critical.
Actions	Inspect, Lubricate, Replace Part, Continue Operation.
Reward	Positive reward for improved health/uptime; negative reward for unnecessary maintenance; large negative reward for failure.
Transition dynamics	After an action, the machine may remain in the same state, improve, deteriorate, or fail. Transition probabilities are defined for each state-action pair.

This is an MDP because the current machine-health state, selected maintenance action, immediate reward and probabilistic transition to the next state form a sequential decision problem.

4(b) Q-Learning for at least 5 episodes

Q-Learning uses: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$. For illustration, $\alpha = 0.5$ and $\gamma = 0.9$, starting with $Q = 0$.

Iteration	State-action	r	max $Q(s',a')$	Old Q	New Q
1	S1-Inspect	5	4	0	4.3
1	S2-Lubricate	8	6	0	6.7
1	S3-Replace Part	15	0	0	7.5
2	S1-Inspect	5	6	4.3	5.85
2	S2-Lubricate	8	7.5	6.7	7.725
2	S3-Replace Part	15	0	7.5	11.25

Example: S1-Inspect in iteration 1: $Q = 0 + 0.5[5 + 0.9(4) - 0] = 4.3$. The agent should be trained for at least five machine episodes as required; in practice, many more episodes are normally needed.

Convergence criterion: stop when the maximum absolute Q-table change over a sufficiently long evaluation window is below a small threshold such as 0.01 and the learned policy remains unchanged across consecutive evaluation episodes.

4(c) Final learned policy and ethical considerations

Machine state	Illustrative best action	Reason
S0 – Healthy	Continue Operation	No immediate maintenance is required; monitor condition.
S1 – Warning	Inspect	Collect more information and identify the developing fault.
S2 – High Risk	Lubricate / Inspect	Take preventive action before the condition becomes critical.
S3 – Critical	Replace Part	Prioritise safety and prevent catastrophic failure.

Ethical considerations: worker safety must take priority over cost or uptime; cost trade-offs must not delay essential maintenance; recommendations should be explainable; human oversight should remain available for high-risk decisions; bias and data quality should be monitored; and the RL agent should be validated in simulation and controlled environments before real-machine deployment.

Note: The assessment document identifies Bhuvaneshwaran J (Register No. 192511402) and gives the manufacturing scenario and required tasks, but it does not contain the underlying sensor dataset. Therefore, numerical metrics, impurity values, Q-table values and feature rankings above are illustrative examples rather than measured results from a real dataset.